# Micro III Notes

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## General Equilibrium

### Def "what resources available?"

1. $L$ goods, $l = 1, ..., L$ (finite).

i. consumption: apples, leisure time, location, etc.
ii. inputs: labor, land, capital
iii. dated goods: apple today vs apple tomorrow
iv. contingent goods: umbrella w rain vs umbrella w/o rain

2. Initial (total) Endowment: $\omega = (\omega_1, ..., \omega_L) \in \mathbb{R}^L_+$
3. Production

ii. Production set $Y \subseteq \mathbb{R}_L$.
iii. Call $y \in \mathbb{R}^L$ a "feasible production plan" if $y \in Y$.

4. Feasibility Set/Resource Set: $R = \{x \in \mathbb{R}^L : x = \omega + y \text{ for some } y \in Y\}$.

### Def "Properties on $Y$"

1. $Y$ is closed and nonempty
2. Possible to do nothing: $(0, ..., 0) = \vec{0} \in Y$
3. No free lunch: $Y \cap \mathbb{R}^L_+ = \emptyset$
4. Free disposal: $y \in Y, \hat{y} \leq y \Rightarrow \hat{y} \in Y$

### Def "feasible"

$x$ is *feasible* if:

1. $x_i \in X_i, \forall i = 1, ..., I$ and
2. $\sum_{i=1}^I x_i = R$ (i.e., $\exists y \in Y : \sum_{i=1}^I x_i = \omega + y$)

### Def "Pareto Efficient"

A feasible consumption plan, $x$ is *Pareto Efficient* (Pareto Optimal) if there does not exist any other feasible $\hat{x}$ such that

1. $\hat{x_i} \succsim_i x_i \forall i = 1, 2, ..., I$ and
2. $\hat{x_i} \succ_i x_i$ for at least one $i \in \{1, ..., I\}$.

If such $\hat{x}$ exists, we say $\hat{x}$ *Pareto-dominates* $x$.

### Remark

1. firms are not people (agents).
2. Pareto efficiency is not about equity.

**Def "algorithm"**

Steps:

1. compute $R$ from $\omega$ and $Y$ to figure out feasible consumption plans.
2. $U_i = \{u_i(x_i) : x \text{ is feasible}\}, \forall i = 1, ..., I.$ (set of all possible utility levels)
3. Fix $\bar{u_i} \in U_i, i = 2, ..., I.$

a. $\max_x u_1(x_1)$ subject to $u_i(x_i) \geq \bar{u_i}, i = 2, ..., I.$

4. Change values: $u_i \in U_i, i = 2, ..., I$ and repeat step 3.

**Remark: the algorithm is not sufficient UNLESS**

1. $u_i$ is continuous and strongly monotone, and
2. $Y = \{y \in \mathbb{R}^L : \vec{y} \leq \vec{0}\}.$

**Prop "Pareto $\Rightarrow x^*$ solution"**

If $x^*$ is Pareto efficient, then $x^*$ solves step 3 above for some $\bar{u_2}, ..., \bar{u_I}, \bar{u_i} \in U_i \forall i = 2, ..., I.$

# Walrasian Equilibrium

**Def "Primitives"**

1. Individual firms
   - $j = 1, ..., J, Y_j \in \mathbb{R}^L$ such that
   - $$Y = \sum_{j=1}^{J} Y_j = \{y \in \mathbb{R}^L : \exists y_j \in Y_j, \forall j = 1, ..., J \text{ s.t. } y = \sum_{j=1}^{J} y_j\}$$
2. Individual endowment
   - $\omega_i \in \mathbb{R}^L_+, \forall i = 1, ..., I$ such that $\omega = \sum_{i=1}^{I} \omega_i$ (previously we had only $\omega$ and not $\omega_i$)
3. Consumer $i$ share in firm $j$'s profits
   - $\underbrace{\theta_i^j \in [0, 1]}_{\text{share of firm j's profits to consumer i for consumption}}$ $, \forall j$ such that $\sum_{i=1}^{I} \theta_i^j = 1$ (consumers own 100% of firms)

**Def "Allocation"**

An allocation is

$$( \underbrace{x_1, ..., x_I}_{\text{consumption plans}} , \underbrace{y_1, ..., y_J}_{\text{production plans}} ) \subseteq \mathbb{R}^{IL+JL}, \quad \underbrace{\forall i, x_i \in X_i}_{\text{consumption is feasible}} , \underbrace{\forall j, y_j \in Y_j}_{\text{production is feasible}} .$$

**Def "Walrasian Equilibrium"**

$p^*, (x^*, y^*)$ are a Walrasian Equilibrium if the following are satisfied:

1. Profit Maximization: $p^* \cdot y_j^* \geq p^* \cdot y_j, \forall y_j \in Y_j, \forall j = 1, ..., J$
2. Preference maximization: $\underbrace{p^* \cdot x_i^*}_{\text{expenditure}} \leq \underbrace{p^* \cdot w_i}_{\text{endowment income}} + \underbrace{\sum_{j=1}^{J} \theta_i^j (p^* \cdot y_{j,l}^*)}_{\text{income from profits}} \ \forall y_j \in Y_j, \forall i = 1, ..., I$
3. Market clearing: $\sum_{i=1}^{I} x_{i,l}^* = \sum_{i=1}^{I} \omega_{i,l}^* + \sum_{j=1}^{J} y_{j,l}^*, \forall l = 1, ..., l.$

**Obs 1 "normalizing prices"**

If $(p^*, x^*, y^*)$ is a Walrasian Equilibrium, then $(\lambda p^*, x^*, y^*)$ is a Walrasian Equilibrium $\forall \lambda > 0$

**Obs 2 "Walras' law"**

If $\succsim_i$ is locally nonsatiated and $(p^*, x^*, y^*)$ satisfies profit maximization then

$$\underbrace{p^* \cdot \sum_{i=1}^{I} x_i^*}_{\text{agg demand}} = \underbrace{p^* \cdot \sum_{i=1}^{I} \omega_i + p^* \cdot \sum_{j=1}^{J} \overbrace{\sum_{i=1}^{I} \theta_i^j}^{=1} (y_j^*)}_{\text{agg supply}} .$$

**Obs 3 "markets clear for all but one, then markets clear for all"**

If 1. $\succsim_i$ is locally nonsatiated, 2. $(p^*, x^*, y^*)$ satisfies profit maximization, 3. $p_L^* \neq 0$, and 4. $\sum_{i=1}^{I} x_{i,l}^* = \sum_{i=1}^{I} \omega_{i,l} + \sum_{j=1}^{J} (y_{j,l}^*), \forall l = 1, ..., L-1$,

Then $\sum_{i=1}^{I} x_{i,L}^* = \sum_{i=1}^{I} \omega_{i,L} + \sum_{j=1}^{J} (y_{j,L}^*)$

In words: if market clearing holds for all other goods, and the price of the last good is nonzero, then I have market clearing for the last good.

**Thm "First Welfare Theorem (WE $\Rightarrow$ Pareto-efficient)"**

Suppose $\succsim_i$ is locally nonsatiated $\forall i = 1, ..., I$ and $(p^*, x^*, y^*)$ is a Walrasian Equilibrium. Then $(x^*, y^*)$ is Pareto-efficient.

**Remark: "Implicit assumptions First Welfare Thm"**

1. firms are price-taking
2. no externalities
3. markets are complete (less relevant to this course, but an important assumption in finance theory.)

**Def "system of transfers"**

A system of transfers, $T$ is a vector $T = (T_1, ..., T_I) \in \mathbb{R}^I$ such that $\sum_{i=1}^{I} T_i = 0$.

Note:

- $T_i > 0 \Rightarrow$ "receiving transfer"
- $T_i < 0 \Rightarrow$ "paying transfer (tax)"

**Def "WE with transfers"**

$(p^*, x^*, y^*)$ is a Walrasian equilibrium with transfers $T$ if it satisfies 1. Profit maximization (no change here), 2. Preference maximization: $x_i^*$ maximizes $\succsim_i$ (represented by $u_i$ subject to $p^* \cdot x_i^* \leq p^* \omega_i + \sum_{j=1}^{J} \theta_i^j p^* \cdot y_j^* + T_i$), and 3. Market clearing (no change here either).

Note: we always add $T_i$ even if $T_i < 0$.

**Thm "First Welfare Theorem with transfers"**

Suppose $T$ is a system of transfers, $\succsim_i$ is locally nonsatiated $\forall i = 1, ..., I$ and $(p^*, x^*, y^*)$ is a Walrasian Equilibrium with T. Then $(x^*, y^*)$ is Pareto-efficient.

**Thm "Second Welfare Theorem (PE $\Rightarrow$ WE)"**

Under assumptions 1-4, if $(x^*, y^*)$ is Pareto-Efficient then there exists a system of transfers $T$ and $p^* \neq \vec{0}$ such that $(p^*, x^*, y^*)$ is a Walrasian Equilibrium with transfers $T$.

Assumptions: 1. $\forall i, X_i, \succsim_i$ satisfies * convexity: $x_i \succsim x_i' \Rightarrow \lambda x_i + (1-\lambda)x_i' \succsim x_i', \forall \lambda \in (0,1)$, and * something slightly weaker than strict convexity: $\forall x_i, x_i', \hat{x}_i \in X_i : [x_i \succ_i \hat{x}_i, x_i' \succ_i \hat{x}_i] \Rightarrow \lambda x_i + (1-\lambda)x_i' \succ_i \hat{x}_i, \forall \lambda \in (0,1)$.) 2. $Y_j$ is convex $\forall j = 1, ..., J$. 3. $\succsim_i$ is locally nonsatiated 4. $\forall i$ * $\succsim_i$ is continuous, and * $X_i \in \mathbb{R}^L_{++}$. This one is VERY STRONG – means that EVERY individual has a nonzero amount of EVERY good.

Note: assumption 4 is weaker in MWG and other places, but we will always use this version in this class.

# Existence of Walrasian Equilibria

**Def "set of price vectors"**

$\Delta = \{p \in \mathbb{R}^L : \forall l, p_l \geq 0 \text{ and } \sum_{l=1}^L p_l = 1\}$

**Def "excess demand function"**

$\varphi : \Delta \to \mathbb{R}^L$ such that $\forall p \in \Delta$:

$$\varphi(p) = \sum_{i=1}^{I} x_i^*(p) - \sum_{i=1}^{I} \omega_i - \sum_{j=1}^{J} y_j^*(p)$$

is called the excess demand function.

$\varphi(p) = \vec{0} \Rightarrow$ WE exists. And, under some assumptions we can get the converse:

WE exists $\Rightarrow \varphi(p) = \vec{0}$.

**Note "Implicit assumptions of excess demand fn"**

1. We can focus on $p \in \Delta$. That is, in WE, $p_l \geq 0 \forall l$ and for at least one good $l, p_l > 0$.
2. There is a unique solution to the profit and preference maximizing problems (i.e., $\varphi$ is a function, not a correspondence.). This one isn't really a problem though because we can always generalize later and use Kakutani's instead of Brouwer's fixed point theorem.
3. Excess demand for any good is finite (even for \$0-priced goods).

**Thm "sufficient conditions on $\varphi$"**

Suppose $\varphi : \Delta \to \mathbb{R}^L$ is well-defined (i.e., satisfies implicit assumptions 1-3) and satisfies

1. $\varphi$ is continuous,
2. $\varphi$ satisfies Walras' law ($p \cdot \varphi(p) = 0, \forall p \in \Delta$), and
3. if $p_l = 0$ for some $l$ then $\varphi(p_l) > 0, \forall p \in \Delta$,

Then, there exists at least one $p \in \Delta$ such that $\varphi(p) = \vec{0}$

Note (exercise): $f(p) = p \iff \varphi(p) = 0$

**Thm "sufficient conditions on primitives"**

Suppose $\forall i, \forall j$:

1. $X_i \in [0, m]^L$ where $m$ is "large" (i.e, $m = \max_{l=1,...,L}\{\max_{(x,y)}\{x_l\}\} + 1$)
2. $\omega_{i,l} > 0, \forall l, i$
3. $\succsim_i$ is continuous, strongly monotone, and strictly convex, and
4. $\vec{0} \in Y_j, Y_j$ is convex and satisfies $\forall y_j, y_j' : y_j \neq y_j' \Rightarrow \exists \hat{y}_j \in Y_j$ such that $\hat{y}_{j,l} > \lambda \hat{y}_{j,l} + (1-\lambda)\hat{y}_{j,l}', \forall l$.

Then, $\varphi$ is well-defined and satisfies 1-3 of Thm "sufficient conditions on $\varphi$".

## Misc

PET economy = Production, Endowments, Technology

**Term "Quasilinear economy"**

$$U(x_i, m) = U(x_i) + m$$